
ANGULAR MOMENTUM IN NONLINEAR OPTICS

In 1993 Basistiy *et al.* (7.1) examined the interaction of beams containing optical vortices with nonlinear materials and with computer generated holograms. In each case additional vortices, of either the same or opposite handedness, were found to appear in the beam. It was noted that when frequency doubled a beam with a first order screw dislocation, that is $l=1$, acquired an additional screw dislocation.

The first work in which the role of orbital angular momentum was explicitly considered in a nonlinear process was that of Dholakia *et al.* (7.2). They demonstrated that the frequency doubling of an LG mode gave rise to another LG beam with its azimuthal mode index doubled. They were able to explain this in terms of the phase matching conditions in the crystal which dictated that the Poynting vector should maintain the same spiral trajectory through the crystal. The corresponding doubling of the orbital angular momentum per output photon meant that the total orbital angular momentum was conserved within the light field. The same group generalised their work to include LG modes with different radial as well as different azimuthal mode indices (7.3, **Paper 7.1**). The frequency doubled LG mode with $p=0$ is another LG mode with index $2l$ and $p=0$. But, for $p>0$ the amplitude distribution of the resulting mode, the square of that of the input mode, is a superposition of LG beams of different mode order. Although still conserving orbital angular momentum, the beam has a form that changes upon propagation and which only reproduces the beam waist distribution in the far-field.

Second harmonic generation is a special case of sum frequency mixing in which two input fields of frequencies ω_1 and ω_2 produce a third field $\omega_3 = \omega_1 + \omega_2 = 2\omega_1$. Berzanskis *et al.* (7.4) experimented with sum frequency generation for beams with a variety of l -values between 1 and 3 and interpreted the results in terms of topological charge. Conservation of orbital angular momentum would dictate that $l_3 = l_1 + l_2$.

For high incident intensities, the possibility of additional non-linear processes needs to be considered. For sufficient intensity, both second- and third-order nonlinearities can give rise to self-defocusing and self-focusing leading to fragmentation of the beam. In 1992 Swartzlander and Law demonstrated that, in a self-defocusing medium, an optical vortex could be stable and be used to guide a probe beam (7.5). In 1997, Torner and Petrov (7.6) predicted that in a second-order material an intense input LG_0^1 beam would, because of self-focusing, break up into three. Firth and Skryabin (7.7, **Paper 7.2**) later predicted that subject to a second- or third-order non-linearity, a single annular ring can break up into $2|l|+1$ or $2|l|$ fragments respectively. Because of their soliton properties, these fragments can retain their transverse localisation. The fragments resemble Newtonian

particles flying off at a tangent to the initial ring; a vivid demonstration of the conservation of angular momentum. Under different conditions, Soljačić and Segev (7.8, **Paper 7.3**) have predicted that the individual fragments can rotate in a manner akin to a rigid body.

Second harmonic generation and sum frequency mixing are examples of three-wave interactions, where the phase of the two input fields define the phase of the third output beam. Parametric down conversion is also a three-wave interaction and again there is a fixed phase relation between the input pump and output signal and idler fields. However, neither the signal nor idler have an externally set phase and it is their phase sum that is defined by the pump. Classically, the signal and the idler fields are both spatially incoherent (7.9) and neither, therefore, have phasefronts with a well defined azimuthal index. It follows that for classical fields neither signal nor idler beams have a unique orbital angular momentum (7.10). However, at the single-photon level, the phase relationship does imply a fixed relationship between signal and idler beams. Recently, Mair *et al.* (7.11, **Paper 8.1**) have shown that although the orbital angular momentum of the signal and idler photons have a range of values, each photon pair conserves the orbital angular momentum; that is, $l_3 = l_1 + l_2$.

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Second-harmonic generation and the conservation of orbital angular momentum with high-order Laguerre-Gaussian modes

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(Received 16 June 1997)

Laguerre-Gaussian modes of various order are frequency doubled. The azimuthal phase structure of the second-harmonic light is measured directly by interfering the beam with its mirror image. We show that the orbital angular momentum per photon is doubled, so conserving the orbital angular momentum in the light beam. The frequency-doubled output beam is shown to have a Gegenbauer-Gaussian amplitude distribution at the beam waist. The beam can be described as a summation of Laguerre-Gaussian modes that interfere so that it changes form with propagation, but the distribution at the beam waist is reproduced in the far field. [S1050-2947(97)03311-8]

PACS number(s): 42.65.Ky, 42.60.Jf

It was predicted in 1992 that monochromatic beams with an azimuthal phase term $e^{il\phi}$, of which Laguerre-Gaussian laser modes are an example, have a well-defined orbital angular momentum of $l\hbar$ per photon [1]. This orbital angular momentum is associated with the azimuthal component of the Poynting vector [2] and is quite distinct from the spin angular momentum associated with circular polarization. The transfer of this orbital angular momentum to a microscopic particle has been demonstrated recently [3]. Microscopic particles held within a Laguerre-Gaussian laser beam have been rotated and the orbital angular momentum quantified by a comparison of this rotation to that induced by the spin angular momentum. The results confirm that the orbital angular momentum is $l\hbar$ per photon [4,5]. Other work has explored theoretically the interaction of these beams with atomic systems [6,7].

Laguerre-Gaussian modes [8] are characterized by two indices l and p , where l is the number of 2π cycles in phase around the circumference and $p+1$ the number of radial nodes. The amplitude u_p^l of such a mode in cylindrical coordinates is given by

$$u_p^l(r, \phi, z) \propto \exp(-ikr^2/2R) \exp(-r^2/w^2) \times \exp[-i(2p+l+1)\psi] \times \exp(-il\phi) (-1)^p (r\sqrt{2}/w)^l L_p^l(2r^2/w^2), \quad (1)$$

where r is the distance from the beam axis, ϕ the azimuthal angle, z the distance from the beam waist, k the wave number of the light, w the radius for which the Gaussian term falls to $1/e$ of its on-axis value, z_r is the Rayleigh range, $L_p^l(x)$ an associated Laguerre polynomial, and $(2p+l+1)\psi$ is the Gouy phase, where $\psi = \arctan(z/z_r)$.

In 1996 we reported the mode transformation that occurs when a $p=0$ Laguerre-Gaussian mode is frequency doubled [9]. Modes with $p=0$ have a single-annular-ring intensity distribution. They frequency double to give a pure Laguerre-Gaussian mode also with $p=0$, but with an azimuthal index of $2l$, that is, with twice the orbital angular momentum per photon. This mode transformation is readily understood in terms of the spiraling of the Poynting vector, which has the

same form for both the fundamental and second-harmonic beams, and was shown to be consistent with conservation of orbital angular momentum within the light beams. In our earlier work the azimuthal phase structure of the frequency-doubled mode was not measured directly. It was deduced by converting the second-harmonic mode into the corresponding Hermite-Gaussian mode by means of a cylindrical lens mode converter [10].

In this paper we confirm our earlier results by the direct measurement of the azimuthal phase structure of the frequency-doubled beams and extend the results to include frequency doubling of the multiringed, $p>0$, Laguerre-Gaussian modes. We show that this results in beams that possess a well-defined orbital angular momentum, but are not simple Laguerre-Gaussian modes.

At the beam waist, $z=0$, the amplitude of a Laguerre-Gaussian mode simplifies to

$$u_p^l(r, \phi, z=0) \propto e^{-r^2/w_0^2} e^{-il\phi} (-1)^p (r\sqrt{2}/w_0)^l L_p^l(2r^2/w_0^2). \quad (2)$$

For $p=0$, the associated Laguerre polynomial is a constant and independent of r . In second-harmonic generation, the amplitude of the frequency-doubled field is proportional to the square of the incident field [11]. It follows that for a $p=0$ mode the second-harmonic beam is also a Laguerre-Gaussian mode that has undergone the following transformations: $k \rightarrow 2k$, frequency doubling; $w_0 \rightarrow w_0/\sqrt{2}$, reduction of the beam waist; $p=0 \rightarrow p=0$, the amplitude distribution remains single ringed; and $l \rightarrow 2l$, the angular momentum per photon is doubled. When $p>0$ the square of the incident field can no longer be described in terms of a single Laguerre-Gaussian mode. However, the azimuthal phase structure is still of the form $\exp(il\phi)$.

The experimental arrangement for generating the frequency-doubled beams, as well as the subsequent analysis of their intensity and phase structure, is shown in Fig. 1. An intracavity cross wire is used to generate a variety of Hermite-Gaussian modes with indices m and n from a diode-pumped Nd:YAG laser (where YAG denotes yttrium aluminum garnet) operating at 1064 nm producing a linearly polarized output power of ≈ 100 mW. Each mode is converted

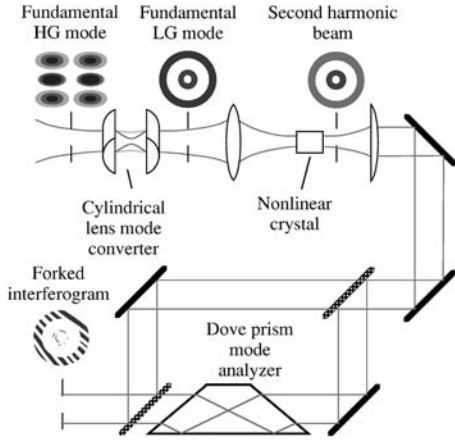


FIG. 1. Experimental apparatus for the generation and analysis of the second-harmonic beams.

into the corresponding Laguerre-Gaussian mode by means of a cylindrical lens mode converter, giving the transformations $l = m - n$ and $p = \min(m, n)$ [10]. The Laguerre-Gaussian modes are then frequency doubled using a 10-mm-long crystal of potassium titanyl phosphate (KTP), angle tuned to give phase matching for the second harmonic at 532 nm. The efficiency of the process is maximized by focusing the incident beam such that its Rayleigh range is comparable to the length of the crystal [12]. Filters placed after the crystal allow either the fundamental or the second-harmonic beam to be selected and imaged onto a charge coupled device array detector. The azimuthal phase structure of the beams can be measured directly using a mode analyzer [13] based on a Dove prism, which allows the interference pattern between the beam and its own mirror image to be obtained. The azimuthal phase component of the beam gives rise to forked interference fringes and the l index of the beam can be inferred directly from the number of fringes on either side of the fork by dividing the number of additional fringes by 2.

Figure 2 shows the forked interferograms obtained for a variety of fundamental Laguerre-Gaussian beams and their second-harmonic counterparts. As in our previous work [9], these results confirm that the azimuthal index l of the beam is doubled in the second-harmonic process. However, here the azimuthal phase has been measured directly and we observe that the doubling of l in the second-harmonic process holds for Laguerre-Gaussian modes of any order of l and p . Again, this is consistent with the conservation of orbital angular momentum within the light beams.

The less than perfect mode converter introduces residual astigmatism into the beam, which manifests itself as a slight ellipticity in the observed images. Corrected radial profiles for the fundamental and second-harmonic beams are obtained by scaling the images to make them symmetrical and averaging the profiles over 80 azimuths. These profiles are then fitted to the predicted field distributions of a Laguerre-Gaussian and its square, respectively, with the amplitude and beam diameter as the fit variables. Figure 3 shows that the corrected profiles are in good agreement with those predicted. The square of a Laguerre polynomial is a Gegenbauer

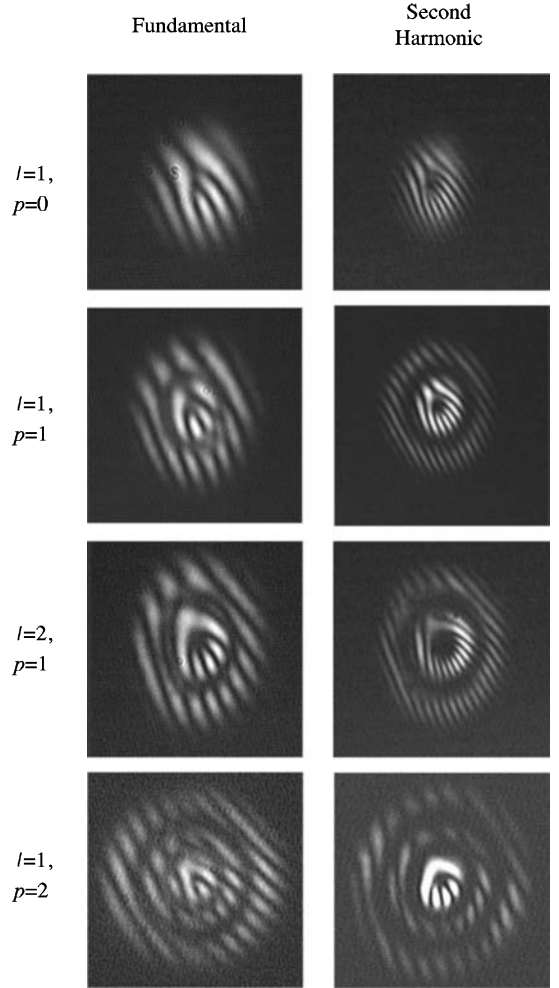


FIG. 2. Forked interferograms derived from a variety of Laguerre-Gaussian beams and their second-harmonic counterparts.

polynomial; consequently, we refer to the amplitude distribution of a frequency-doubled Laguerre-Gaussian mode at the beam waist as a Gegenbauer-Gaussian distribution.

Pure Laguerre-Gaussian modes propagate without changing their form, with a beam divergence dictated by the size of the beam waist and the corresponding Rayleigh range. As discussed, the second-harmonic beam for $p > 0$ can no longer be described as a simple Laguerre-Gaussian mode. Insight as to how it can be described can, however, be gained from the experimental evidence: We find that the distribution of intensity in the far field is the same as that in the plane of the nonlinear process at $z = 0$. For a monochromatic beam the far-field amplitude distribution is simply the Fourier transform of the distribution at the beam waist [14]. When the Fourier transform of light consisting of the square of Laguerre-Gaussians at z is taken, it is found that the resulting distribution can be described as a superposition of a number

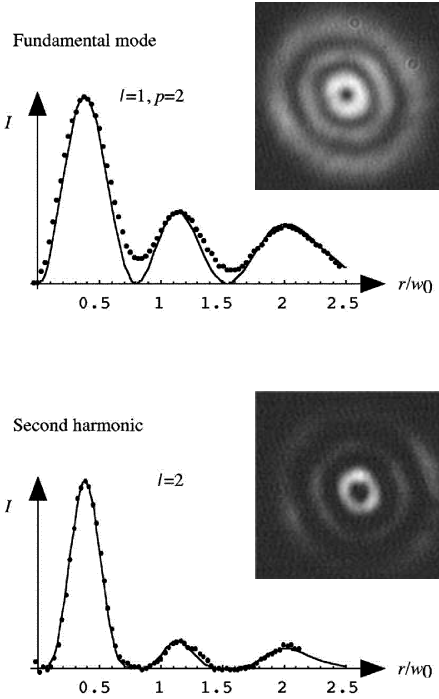


FIG. 3. Corrected radial intensity profiles for the fundamental and second-harmonic beams compared with theory, together with photographs of the observed images recorded in the plane of the frequency-doubling crystal.

of Laguerre-Gaussian modes all with the same index $2l$ but with $p^{(2\omega)} = 0, 2, \dots, 2p$. For $z=0$, this summation reduces to the square of a Laguerre-Gaussian of the same form as in Eq. (2).

Just as a Laguerre-Gaussian mode is a solution of the paraxial Helmholtz equation, so, too, is any sum of Laguerre-Gaussian modes, and as the second-harmonic beam propagates, interference occurs between the constituent modes. Although the modes all have the same Rayleigh range, the p indices give rise to a differing Gouy phase shift between the modes, which leads to an intensity distribution that changes form with propagation. It is only in the far field, where all the Gouy phase shifts differ by multiples of 2π , that the Gegenbauer-Gaussian distribution is reproduced. We see that the behavior of the $p=0$ modes analyzed in our previous paper [9] is simply a limiting case of the general behavior; when $p=0$ there is only one Laguerre polynomial involved and so only one Laguerre-Gaussian distribution.

Rather than investigate the explicit summation of Laguerre-Gaussian modes to determine the beam distribution as it propagates, we have implemented an algorithm based on the Fourier expansion of the beam in k space into a series of plane waves [15]. The propagation of each plane-wave component to a subsequent plane results in a well-defined change in phase. After propagation a summation of the individual plane-wave components followed by an inverse transform gives the new phase and amplitude distribution of the

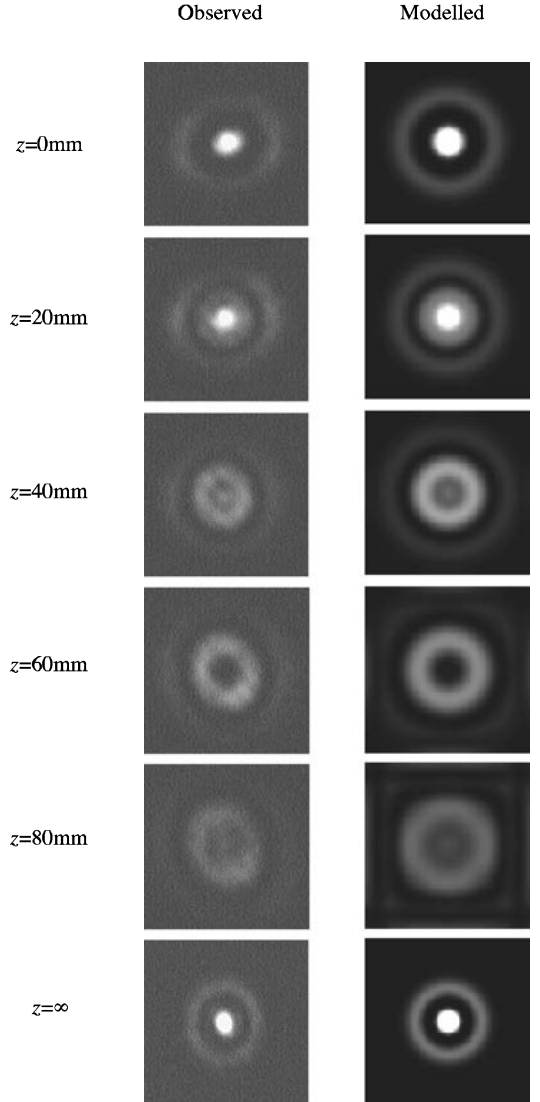


FIG. 4. Observed and modeled intensity distribution of a frequency-doubled Laguerre-Gaussian ($p=1, l=0$) mode as it propagates from the beam waist.

beam. The algorithm is limited only by the array size of the Fourier transforms, which restricts the maximum allowed divergence and physical diameter of the beam. Figure 4 shows the observed and predicted intensity distributions for a frequency-doubled Laguerre-Gaussian mode with $p=1$ and $l=0$ as it propagates from the beam waist to the far field. We see that the intensity distribution of the second-harmonic beam reproduces itself in the far field, but varies at all intermediate positions. We observe similar behavior for all frequency-doubled Laguerre-Gaussian modes with $p>0$.

In this paper we extend our earlier work on the frequency doubling of Laguerre-Gaussian modes to $p>0$ and show that

the previously investigated case of $p=0$ is a limiting case of the general behavior. The azimuthal phase of the frequency-doubled modes is measured directly and the azimuthal phase index is shown to become doubled for all modes. This corresponds to a doubling of the orbital angular momentum per photon during the second-harmonic generation process. The second-harmonic amplitude distribution may be described as a Gegenbauer-Gaussian beam that, while reproducing its intensity distribution in the far field, varies its distribution at all intermediate positions. Nevertheless, the frequency-doubled beams are still well defined and symmetrical about the beam axis. The nonlinearity of the second-harmonic generation

process means that these beams have a radial intensity distribution in which a higher proportion of the energy is closer to the beam axis than in Laguerre-Gaussian modes. Such beams therefore allow the use of lower-aperture optical components in experiments involving the orbital angular momentum of light.

This work was supported by EPSRC. K. D. acknowledges the financial support of the Royal Society of Edinburgh. M. J. P. would like to thank the Royal Society (London) for financial support. It is a pleasure to express our thanks to Stephen M. Barnett for his analytic solution of the relevant Fourier transform.

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Optical Solitons Carrying Orbital Angular Momentum

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(Received 10 April 1997; revised manuscript received 1 July 1997)

We predict a new kind of ring-profile solitary wave in nonlinear optical media, with finite orbital angular momentum. During propagation these fragment into fundamental solitons. Like free Newtonian particles, these fly off tangential to the ring, vividly demonstrating conservation of orbital angular momentum in soliton motion. [S0031-9007(97)04148-3]

PACS numbers: 42.65.Tg, 03.40.Kf, 42.65.Ky

Solitons are important in many branches of science [1]. They are nonlinear waves which possess several mechanical and other attributes more commonly associated with particles. This analogy is usually developed in relation to their response to external “forces,” i.e., in the context of linear momentum; see, e.g., [2]. Here we describe phenomena which are neatly interpreted as solitons carrying *orbital angular* momentum, arising from motion oblique to the centroid of the system.

These solitons are produced in the fragmentation of a new kind of “doughnut soliton.” They have well-defined angular momentum which is transformed into orbital angular momentum of the daughter solitons. Below we demonstrate this phenomenon in two rather different nonlinear optical systems. The close correspondence of the dynamics in the two cases lead us to believe that this scenario should be quite general in solitonic systems with enough dimensions to exhibit angular momentum effects.

Our first model, $\chi^{(3)}$, describes a beam propagating in a saturable self-focusing medium. In the paraxial approximation the evolution in z of the field envelope $E_1(x, y, z)$ obeys the following dimensionless equation [3]

$$i\partial_z E_1 + \frac{1}{2} \tilde{\nabla}_\perp^2 E_1 + E_1 |E_1|^2 / (1 + \alpha |E_1|^2) = 0. \quad (1)$$

Here $\tilde{\nabla}_\perp = \tilde{i}\partial_x + \tilde{j}\partial_y$. For a pure Kerr medium ($\alpha = 0$) this equation is the well-known nonlinear Schrödinger equation (NLS). With the y -dimension suppressed (1D case) the NLS is integrable, with exact solutions—solitons—of sech profile. In 2D it has solitonlike solutions which are unstable, collapsing to a singularity [3]. Saturation, described by a finite positive α , prevents this collapse [3]. Thus, though it is not integrable, Eq. (1) possesses stable solitary wave solutions, localized in 2D, which we will term “solitons.”

Our second model, $\chi^{(2)}$, is physically different, corresponding to the coupled propagation of an optical field and its second harmonic in a quadratically nonlinear medium. Their field envelopes E_1 and E_2 can be de-

scribed by the following system of rescaled equations [4]:

$$\begin{aligned} i\partial_z E_1 + \frac{1}{2} \tilde{\nabla}_\perp^2 E_1 + E_1^* E_2 &= 0, \\ i\partial_z E_2 + \frac{1}{4} \tilde{\nabla}_\perp^2 E_2 + \frac{1}{2} E_1^2 &= \beta E_2. \end{aligned} \quad (2)$$

A recent review [5] provides a good link to experimental parameters, and to complications such as field polarization and walk-off which we neglect here. The parameter β is the phase mismatch. If β is large enough, it can be considered to dominate the derivative terms, then solving for E_2 and substituting into the equation for E_1 gives the NLS. Again Eqs. (2) are not integrable, but stable solitary solutions persist even for small β , far from the NLS limit. We present results for $\beta = 0$ in the following: the phenomena we describe are not very sensitive to β . These solitons are now well known both experimentally and theoretically; see [4–6] and *op cit*.

All quantities in Eqs. (1) and (2) are dimensionless, and these scaled units are used throughout the text and in the figures. Note that both equations have a Galilean invariance, and so the 2D spatial soliton centered on $(x = 0, y = 0)$ generates a family of equivalent solitons which “move” with constant velocity in the (x, y) plane as the beam propagates. It is such moving solitons which are important in the following.

Both models are Hamiltonian, and possess phase, translational, and rotational symmetries. As a consequence, both conserve the energy integral Q , transverse momentum $\vec{P}$, and transverse angular momentum $\vec{L}$, defined as follows: $Q = \int dxdy (|E_1|^2 + 2|E_2|^2)$, $\vec{P} = \int dxdy \vec{p} = \int dxdy \frac{i}{2} [E_1 (\tilde{\nabla}_\perp \cdot E_1^*) + E_2 (\tilde{\nabla}_\perp \cdot E_2^*) - \text{c.c.}]$, $\vec{L} = \int dxdy \vec{r} \times \vec{p}$, where $\vec{p}$ is the transverse momentum density and $\vec{r}$ is the transverse radius vector. (For the $\chi^{(3)}$ model set E_2 to zero.)

The angular momentum carried by light beams has attracted much recent interest. It has been predicted, and proved experimentally, that Laguerre-Gaussian beams with azimuthal model index l carry orbital angular momentum $l\hbar$ per photon [7]. Frequency doubling such a

beam has been shown [8] to generate a second harmonic with doubled azimuthal mode index $2l$.

We now show that both our models admit nondiffracting solitary wave solutions with finite angular momentum. Such “doughnut solitons” are ringlike solutions of (1) or (2) with intensity independent of z . Equations (1) and (2) are reduced by the substitution $E_m(z, r, \theta) = A_m(r) \exp[i m(\kappa z + l\theta)]$ ($m = 1, 2$) to ordinary differential equations which we solve numerically using finite differences. Here $r = \sqrt{x^2 + y^2}$, θ is the polar angle, κ is a real and free parameter, as is l , which we restrict to integer values to ensure azimuthal periodicity. $A_{1,2}(r)$ are real functions which satisfy zero boundary conditions: $A_{1,2}(0) = A_{1,2}(+\infty) = 0$. For the $\chi^{(3)}$ model, consider $m = 1$ only (here and below).

For each l , $\chi^{(3)}$ doughnut solitons exist for all positive κ (which is required for soliton confinement), while for $\chi^{(2)}$ the existence condition is $\kappa > \max(0, -\beta/2)$. Doughnut solitons obey $\vec{P} = 0$ and $|\vec{L}| = |l|Q$ in both models. Typical spatial profiles are presented in Figs. 1(a) and 1(b).

To investigate doughnut soliton stability under propagation, we initialize (1) or (2) with a doughnut soliton (plus noise) and simulate the subsequent evolution on both Cartesian and polar grids, using split-step algorithms. Both approaches give the same results. As a further check, conservation of the energy, momentum, and angular momentum was monitored during the simulations. We find that these doughnuts usually break up into several solitons, which move off at constant “velocity” (angle to the axis of propagation)

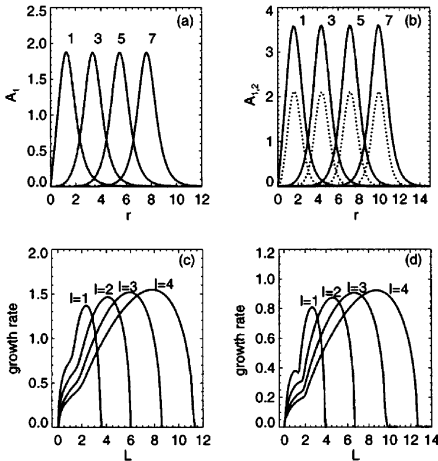


FIG. 1. (a) One-ring solitary wave solutions of Eq. (1): plots of the field amplitude $A_1(r)$ for $\kappa = 1$ and $l = 1, 3, 5, 7$. (b) Corresponding plots of $A_1(r)$ (full) and $A_2(r)$ (dotted) for Eqs. (2). (c) Perturbation growth rate $\text{Re} \tilde{\lambda}$ vs L for different values of l , for the $\chi^{(3)}$ case; $\kappa = 1$, $\alpha = 0.1$. (d) Corresponding plots for the $\chi^{(2)}$ case; $\kappa = 1$, $\beta = 0$.

along paths tangent to the initial ring. The breakup is due to an azimuthal modulational instability. Before discussing the asymptotic motion of the “daughter” solitons, therefore, we outline our doughnut soliton stability analysis.

As regards radial perturbations, the criterion $\partial_\kappa Q > 0$ [9] for stability applies to the doughnut solitons, and is satisfied over most of the existence domain. However, $\partial_\kappa Q > 0$ does not imply stability with respect to azimuthal perturbations [10], which break the cylindrical symmetry. We therefore considered azimuthally perturbed doughnut solitons in the general form $E_m = [A_m(r) + \epsilon_m^+(r, z)e^{iL\theta} + \epsilon_m^-(r, z)e^{-iL\theta}]e^{im(\kappa z + l\theta)}$ ($m = 1, 2$). Substitution into (1) or (2) and linearization results in partial differential equations for $\epsilon_{1,2}^\pm(r, z)$ which we solve numerically using a Crank-Nicholson scheme. We integrate along z until the perturbation growth rate becomes stationary and then average it over a further propagation distance. This procedure gives us an estimate for the real part of the maximally unstable eigenvalue $\tilde{\lambda}$. Alternatively, setting $\epsilon_{1,2}^\pm = [u_{1,2}^\pm(r) + i v_{1,2}^\pm(r)]e^{\lambda z}$ yields a boundary value problem whose spectrum we find by a finite difference method. The two methods give identical results.

Dependencies of the perturbation growth rate $\text{Re} \tilde{\lambda}$ on L for various values of l with other parameters fixed are presented in Figs. 1(c) and 1(d). Physically, L must be an integer to ensure azimuthal periodicity, but it appears in the linearized equations as a real parameter, and extra insight can be gained by studying growth rates for arbitrary positive L . In every case there is a positive growth rate over a range of L values, with a well-defined global maximum for $L = L_{\text{max}}$. We find that azimuthal instability ($L \neq 0$) always dominates radial instability which corresponds to $L = 0$. On propagation, we expect the initially uniform field amplitude around the doughnut to develop N minima and N maxima, where N is the integer closest to L_{max} . As a consequence, the doughnut should break up into N solitons. In the cases illustrated below $N = 2|l|$ for the $\chi^{(3)}$ model and $2|l| + 1$ for $\chi^{(2)}$.

We present results of direct numerical simulation of Eqs. (1) and (2) in Figs. 2 and 3, respectively, for the cases $l = 1, 2, 3$. Figures 2(a)–2(c) and 3(a)–3(c) show the real part of the perturbation eigenmode corresponding to $L = N$ computed from the stability analysis—the real part determines the field amplitude modulation pattern which develops around the initial ring. Figures 2(d)–2(f) and 3(d)–3(f) show the field intensity after propagating far enough for the modulational instability to develop. Each of the N peaks of the eigenmode evolves into an intensity peak (a protosoliton). (For the $\chi^{(2)}$ case the field amplitudes E_1, E_2 are localized in the same region of space due to their nonlinear coupling.)

Figures 2(g)–2(j) and 3(g)–3(j) show the real part of E_1 , indicating that adjacent protosolitons are out of phase (this is less obvious in Fig. 3 because N is odd).

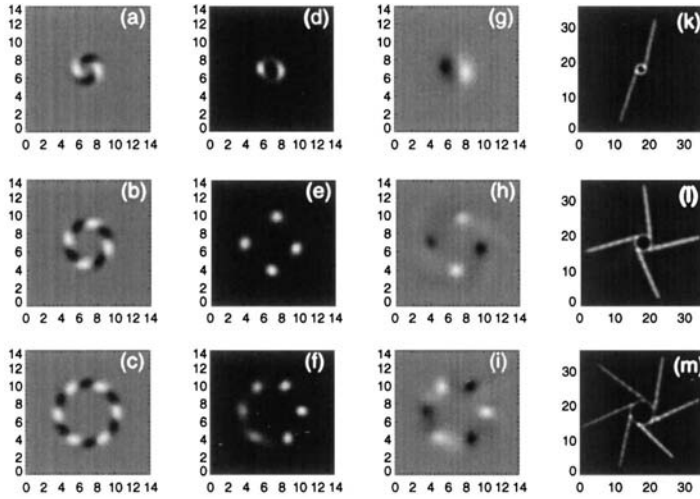


FIG. 2. Azimuthal modulational instability development and soliton trajectories, for $\chi^{(3)}$ case: $\alpha = 0.1$, $\kappa = 1$, with $l = 1, 2, 3$. (a)–(c) Real part of the perturbation field pattern with maximal growth rate. (d)–(f) Numerically computed field intensity $|E_1|^2$ at a point where protosolitons have developed. (g)–(i) Real part of E_1 at the same point, showing relative phases of protosolitons. (k)–(m) Superimposed images of the transverse intensity distribution at different z values, showing soliton trajectories tangential to the initial ring. Note the change of scale.

It has been noted that out-of-phase solitons repel each other, while in-phase attract; see, e.g., Ref. [11]. By symmetry, the resultant force on each protosoliton should thus be *radial* and outward. In Figs. 2(k)–2(m) and 3(k)–3(m) we superimpose a succession of images at different z , to show the daughter soliton trajectories. Far from being radial, these are *tangential* to the initial ring.

Our interpretation is that the intersoliton forces are actually negligible, and that the solitons are behaving

like free Newtonian particles, flying off tangential to the doughnut soliton ring, and carrying away its angular momentum *via* the obliquity of their paths. The “mass” of the n th soliton is its energy Q_n , and its angular momentum is $\vec{r}_n \times \vec{p}_n$, where Q_n and $\vec{p}_n$ are now defined *locally* around the n th soliton’s position $\vec{r}_n$. Using conservation of energy and angular momentum and the Galilean invariance of Eqs. (1) and (2) we calculate the transverse speed of the solitons to be $|\vec{v}| = |l|/R$, where

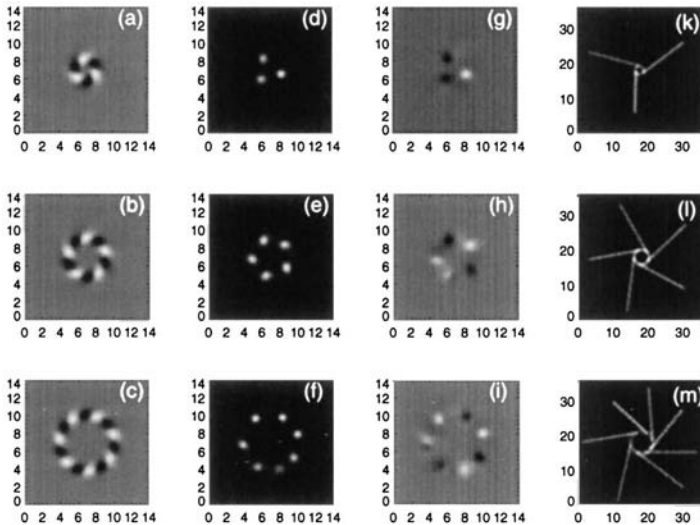


FIG. 3. Same as Fig. 2, but for field E_1 in the $\chi^{(2)}$ model: $\beta = 0$, $\kappa = 2$.

R is the radius of the initial doughnut (the same speed for each daughter soliton, independent of its energy).

This estimate is in good agreement with the numerical results shown in Figs. 2 and 3, with losses (due to nonsoliton “radiation”) at most 10%. Thus, in contrast to recent work [11] in which interacting photorefractive solitons spiraled around each other, here we have nearly free solitons, with dynamics dominated by angular momentum conservation. Interactions forces may play a minor role in partitioning the energy among the protosolitons, but the exponential localization of the daughter solitons means they rapidly cease to interact.

Generalizing from the particular cases displayed in Figs. 2 and 3, we find very similar behavior over a wide range of parameter values. For both models the sign of l , which defines the direction of the angular momentum, fixes the orientation of the soliton trajectories. The number of daughter solitons depends strongly on the azimuthal index l , but relatively weakly on all other parameters.

While we believe that our results are conceptually and pedagogically interesting, independent of direct experimental verification, it is naturally of interest to address the question of experimental observation of these or related phenomena. We already mentioned the work of Shih *et al.* [11], and several other recent papers report experiments which relate in some way to our scheme and are thus generally encouraging, though in each case there are important and interesting differences.

Tikhonenko *et al.* report experiments in rubidium vapor (a saturable self-focusing medium) [12,13]. They observed, and confirmed in simulations, fragmentation of beams with a phase dislocation. They did not consider the possibility of doughnut solitons. Their experimental results were quite strongly affected by the lack of azimuthal symmetry of their input beam, especially in the case $l = 2$ [13]. For the case of $l = 1$ their results [12] are quite similar to ours, with two daughter solitons produced. A side view indicates straight-line trajectories with no evidence of interaction forces leading to spiraling.

In the $\chi^{(2)}$ case, Torner and Petrov [14] recently described numerical simulation for the case where the input was a Laguerre-Gaussian for E_1 with $E_2 = 0$. They concentrated on the case $l = 1$. Typically three solitons were output, broadly in accord with our findings, though for quite different initial conditions. Stationary solutions such as doughnut solitons, we would argue, are the natural starting point for studies of more general input conditions.

Finally, Fuerst *et al.* recently reported an experiment [6] quite close to our model, except that the input is “unrolled” to form an intense line focus. The observed filamentation into up to six stable solitons is encouraging for similar experiments with finite angular momentum.

In conclusion, we have shown the existence of a new kind of ringlike solitary wave, with finite orbital angular momentum. These are unstable on propagation, breaking into filaments which become solitons, whose number is strongly dependent on the input angular momentum. The solitons fly out tangentially from the initial ring, like free Newtonian particles, and their motion is accurately described using Newtonian conservation laws for energy, momentum, and angular momentum. This scenario is rather robust, with broadly similar phenomena occurring over a wide range of parameters, both in quadratically nonlinear and in self-focusing media.

This work was partially supported by EPSRC Grant No. GR/L 27916. D. V. S. acknowledges financial support from ORS award scheme.

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Integer and Fractional Angular Momentum Borne on Self-Trapped Necklace-Ring Beams

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(Received 14 March 2000; revised manuscript received 8 September 2000)

We present self-trapped necklace-ring beams that carry and conserve angular momentum. Such beams can have a fractional ratio of angular momentum to energy, and they exhibit a series of phenomena typically associated with rotation of rigid bodies and centrifugal force effects.

DOI: 10.1103/PhysRevLett.86.420

PACS numbers: 42.65.Tg, 05.45.Yv

Many nonlinear-wave systems can be described by the cubic nonlinear Schrödinger equation (NLSE). Solitons in the $(1 + 1)$ D self-focusing version of this equation are stable, displaying interesting physics and applications, yet solitons of $(2 + 1)$ D NLSE are highly unstable [1]. Recent papers [2,3] have proposed self-trapped $(2 + 1)$ D beams that propagate in a stable fashion in a self-focusing Kerr medium: the necklace-ring beams. Necklace beams are shaped like rings whose thickness w is much smaller than their radius R and whose intensity is azimuthally periodically modulated (Fig. 1). Such beams exhibit stable self-trapped propagation for many (>50) physical diffraction lengths [2,3], even though circular $(2 + 1)$ D solitons are inherently unstable in self-focusing Kerr media [1]. In necklace beams, it is the interaction between the spots that stabilizes the structure as a whole. As shown in [3], an isolated individual spot is highly unstable. Furthermore, removing a single spot from the necklace renders the entire necklace unstable. Necklace beams can be thought of as a superposition of two rings carrying equal but opposite topological charge. For such a superposition to exhibit stable propagation, the thickness of the ring must be significantly smaller than its radius, and the thickness must be larger than the azimuthal period. Moreover, the ring has to propagate as one entity, or else the spots walk off each other. In contrast to necklace beams, a single charge-carrying (ring) beam is highly unstable in any self-focusing medium: In a Kerr material it disintegrates, while in saturable nonlinear media it breaks into a number of solitons that can interact with one another [4] or fly off like free particles [5]. This behavior occurs also in quadratic media [6]. Yet necklace beams stay intact and display stable propagation, in Kerr as well as in saturable self-focusing media, if their parameters are chosen properly [2,3]. The stability of self-trapped necklaces (with properly chosen parameters) is unique in soliton science: A superposition of bound solutions (the rings with equal but opposite charge that make up the necklace) is stable, but its individual constituents are *unstable* [7]. Experiments with self-trapped necklaces have already been reported [8]. Here, we present self-trapped necklace beams that carry angular momentum. It is a rare case of self-trapped scalar bright beams that carry angular momentum [9]. In contrast to all known soli-

tons, the angular momentum borne on such necklaces can be a noninteger multiple of the energy. We demonstrate angular momentum and centrifugal force effects.

Solitons that carry angular momentum have been studied in many systems: vortex solitons [10], 3D spiraling of solitons [11], composite solitons [12], quadratic solitons [13,14], and ring solitons in Kerr [15] and cubic-quintic [9] media. Experiments involving transfer of angular momentum carried by light to other forms of angular momentum have been performed [16]. The angular momentum of optical beams is typically associated with azimuthal phase modulation of $\exp(iM\theta)$ [17], which provides angular velocity to every part of the beam with respect to the beam center. The field of solitons is continuous wherever the amplitude is nonzero, that is, everywhere except for the origin. This is because a field discontinuity where the amplitude is nonzero renders the soliton highly unstable, even in a self-defocusing medium. For this reason of stability, for all vortex solitons [having $\exp(iM\theta)$], M is an integer [14]. For the same reason, for all other forms of single solitons carrying angular momentum, Mu is an integer. In the $(2 + 1)$ D cubic self-focusing NLSE,

$$i \frac{\partial \psi}{\partial z} + \frac{1}{2} \left\{ \frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} \right\} + |\psi|^2 \psi = 0, \quad (1)$$

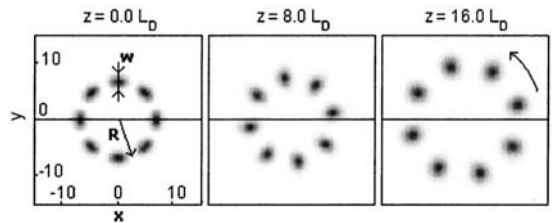


FIG. 1. A rotating necklace with integer L/E . We launch a necklace close to the self-trapped shape. It “breathes” for a short distance, until it reaches the equilibrium shape. This necklace slowly rotates as it propagates. Every necklace slowly expands as it propagates; nevertheless, as seen here, this does not stop the rotation. The input shape is $\sim \psi(r, \theta, z = 0) = \text{sech}(r - 6.83) \cos(4\theta) \exp(i\theta)$. Dark means high intensity.

the energy is $E = \iint |\psi|^2 dx dy$ and the angular momentum is $L\hat{z} = \frac{i}{2} \iint \mathbf{r} \times \{\psi \nabla \psi^* - \psi^* \nabla \psi\} dx dy$. Generically, any beam that can be written as $\psi(r, \theta) = f(r) \exp(iM\theta)$ has $L/E = M$. Since definitions of neither L or E depend on the nonlinearity, this holds for all nonlinearities. Thus, all optical solitons found so far carry an integer L/E . The angular momentum carried by such a beam, when averaged over the number of photons, is exactly $M\hbar$ per photon [17]. If the light is circularly polarized, the total angular momentum is modified by the spin contribution of $\pm\hbar$ per photon.

The fact that L/E is an integer can be intuitively understood by comparison with quantum mechanics (QM). All (paraxial) optical solitons are described by a Schrödinger-type equation. In QM, the solutions of this equation have a quantized angular momentum which is an integer multiple of $\hbar$, and the total probability $\iint |\psi|^2 dx dy$ is normalized to 1. In classical optics $\iint |\psi|^2 dx dy$ is the total power, which is proportional to the average number of photons. The quantization of L in QM resembles the fact that L/E is an integer for all solitons found thus far of a classical (2 + 1)D normalized NLSE. But L/E is an integer for solitons not due to quantization reasons, but because a noninteger L/E typically leads to a field discontinuity where the intensity is nonzero; such a discontinuity is thought to be unstable in self-focusing/defocusing media. Here we find self-trapped structures carrying noninteger L/E yet stably propagating for many diffraction lengths: necklace-ring quasisolitons that carry noninteger per-photon angular momentum.

Consider a necklace beam whose input shape is approximately $\psi(r, \theta, z = 0) = f(r) \cos(\Omega\theta)$ [18], and add to it angular momentum by multiplying it by $\exp(iM\theta)$ (Ω , M integers). As long as M is reasonably smaller than Ω , we find (numerically) that this necklace is stable for more than 50 diffraction lengths L_D [19]. After 50 L_D we reach our computational limits, but it is plausible that the necklaces are stable for much larger distances. Such shapes have $L/E = M$. Since the symmetry between the spots must be preserved in a stable propagation, the angular momentum is manifested by the rotation of the entire necklace as it propagates (Fig. 1). We find numerically that quantization of L/E means that, for a necklace whose parameters are all fixed (except for its M), only certain angular velocities ω are allowed; these ω 's are given by $\approx M/R^2$. Two necklaces that differ only in their radii (have the same M) differ in their ω 's by a squared ratio of their radii. The fact that the allowed L/E 's are quantized shows a connection between solitons and bound states in QM. Both of these systems are described by very similar wave equations and display several similar properties. The fact that some wave quantities that relate solitons and particles are necessarily quantized in optics was not appreciated so far.

Self-trapped necklace beams slowly expand as they propagate [3,4]. The expansion is a consequence of the net radial force exerted on each spot in the necklace. However, even though the necklace slowly expands,

the dynamics is very different (and much slower) than diffractive dynamics: It is uniform and it preserves the shape of the necklace. Once angular momentum is added to a necklace beam, it expands faster. The expansion is still highly dominated by the internal dynamics of the necklace. Nevertheless, for two necklaces that differ only in their M 's, the one with larger M expands noticeably faster. This implies that what we observe is actually a centrifugal force in a solitonic system. Furthermore, as a necklace beam expands, its L and E are conserved, implying that ω (the angular velocity) cannot be conserved. This is similar to a skater on ice: If she extends her hands while rotating, her ω decreases. We observe this with necklace beams (Fig. 2). Analytically, the angular phase has to be conserved; otherwise, because ω is quantized, the phase would discontinuously jump from, say, $\exp(i\theta)$ to $\exp(i2\theta)$, which is not physical in the continuous evolution describing the necklace propagation. Thus, as the necklace expands, ωR^2 is conserved. Our numerics confirm this prediction. One can develop a moment of inertia formulation for this system. The moment of inertia for necklaces is $I \approx ER^2$. Since $L = I\omega$ and E and L are conserved, ω has to go down with R^2 . This is the first prediction of a "skater on ice" effect, which is so obvious in Newtonian mechanics but is unobserved yet in solitonic systems: the slowing down of angular velocity due to conservation of energy and angular momentum.

Given that, in analogy to QM, only integer L/E values are allowed, we recall that there are objects that carry angular momentum (in the form of spin) in multiples of $\hbar/2$ also. However, such spin is an internal degree of freedom and cannot be reproduced as a manifestation of a spatial property of a wave function. Nevertheless, even in QM, the expectation value of angular momentum can be a noninteger multiple of $\hbar$. We build on this idea to construct stable self-trapped beams that carry noninteger L/E . The necklaces described above have $\psi(r, \theta, z = 0) = f(r) \{\exp[i(\Omega + M)\theta] + \exp[-i(\Omega - M)\theta]\}/2$. To create a necklace

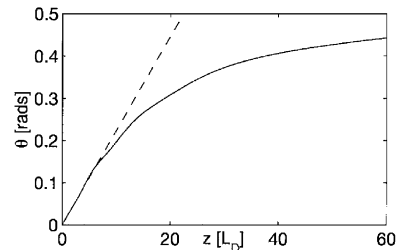


FIG. 2. The rotation angle of the expanding necklace of Fig. 1, as a function of the propagation distance. This particular necklace expands significantly as it propagates. The solid line represents the true instantaneous angle of rotation (measured numerically), whereas the dashed line represents what the instantaneous angle of rotation would have been if the angular velocity were a conserved quantity.

carrying noninteger L/E , we launch $\psi(r, \theta, z = 0) = f(r) \{ \exp[i(\Omega + M)\theta] + \exp[-i(\Omega)\theta] \}/2$, as in Fig. 3. Such a necklace has $L/E = M/2$. For an odd M , this necklace has a noninteger L/E . Its intensity is given by $f^2(r) \{ 1 + \cos[(M + 2\Omega)\theta] \}/2$. In contrast to necklaces that have an even number of spots [3,4], a necklace that carries noninteger L/E has an odd number of spots. Furthermore, in a necklace with an integer (or zero) L/E , adjacent spots are mutually π out of phase (this is why such necklaces expand). This is not the case here since there is an odd number of spots. In order to preserve symmetry between the spots, the angular momentum is manifested in rotation of the necklace, and $\omega = M/(2R^2)$; thus ω is twice slower than for the corresponding necklaces of the previous paragraphs, keeping M and other parameters fixed. Our numerics confirm this prediction, and these necklaces are as stable as the usual necklaces: for many tens of L_D 's. We investigate the stability of these necklaces using the same methods as in [2,3].

Another surprising feature is that, although the probability to find photons is not azimuthally symmetric (hence, the nonuniform azimuthal intensity), the local expectation value of L/E is azimuthally symmetric [20]. That is, we calculate analytically the ratio $L(\theta)/E(\theta)$ at $z = 0$ and find this ratio to be independent of θ , both in the case of the necklace with an even number of pearls (when it equals M) and in the case of a necklace with an odd number of pearls (when it is $M/2$). Because these necklaces rotate as rigid bodies we expect the $L(\theta)/E(\theta)$ not to change significantly during propagation. Thus, in a necklace with an odd number of spots, each photon contributes exactly $M\hbar/2$ the expectation value of the total angular momentum. One might think that a noninteger per-photon angular momentum is because different regions of the beam have different ratios of $L(\theta)/E(\theta)$, but this is not the case; since the shape of each spot is fixed as the necklace propagates, each part of the beam has the same angular velocity with respect to the center of the necklace.

Next, we construct a necklace carrying an arbitrary real per-photon angular momentum. Consider a necklace with $\psi(r, \theta, z = 0) = f(r) \{ a \exp(iM\theta) + b \exp(iN\theta) + c \exp(-iP\theta) + d \exp(-iQ\theta) \}$; it has $L/E = (a^2M + b^2N - c^2P - d^2Q)/(a^2 + b^2 + c^2 + d^2)$, which can take any real value. Not all such necklaces are stable, but

one can construct necklaces that are stable for many L_D 's. Setting $N = P$, M , and Q to have similar values as N and P , b to be similar to c , and $a, d \ll b, c$, the necklace looks like a "usual" necklace, but with its envelope slightly azimuthally modulated. Small azimuthal perturbations do not destabilize necklaces: we find many such necklaces that are stable for more than $20L_D$, which is plenty for experimental observations. In Fig. 4, we show a necklace that has $d = 0$, $N = P = 8$, $M = 15$, $a = 1$, $b = 7$, and $c = 8$. Therefore, $L/E = -35/38$ for this necklace. As shown in Fig. 5, this necklace indeed has a shape similar to a usual necklace, but with a small azimuthal perturbation. It is interesting to note that in these necklaces the angular momentum is not manifested just in rotation of the necklace, but also in circulation of the modulation of the azimuthal envelope (Fig. 4): Neighboring spots exchange energy and perform a circulation of energy around the necklace, and this is the primary means of transporting the angular momentum upon the propagating beam. The reason for this distinctly different behavior of this necklace from the necklaces with integer or $M/2$ L/E values is symmetry. For a necklace with integer or $M/2$ L/E , the symmetry between the spots is conserved. Thus, if a symmetric necklace is to stay stable, the only way the angular momentum can be manifested is the rotation of the necklace as a whole (Figs. 1 and 3). In contrast, for a necklace described in this paragraph, the symmetry between spots is broken. Thus, spots are allowed to exchange energy and thereby carry angular momentum without a significant rotation of the necklace. Indeed, the frame of the necklace appears stationary (Fig. 4), yet the spots circulate the energy in a preferential direction corresponding to the sign and value of L/E .

The necklace of Figs. 4 and 5 has $L(\theta)/E(\theta)$ which depends on θ . Since in this necklace the spots are not

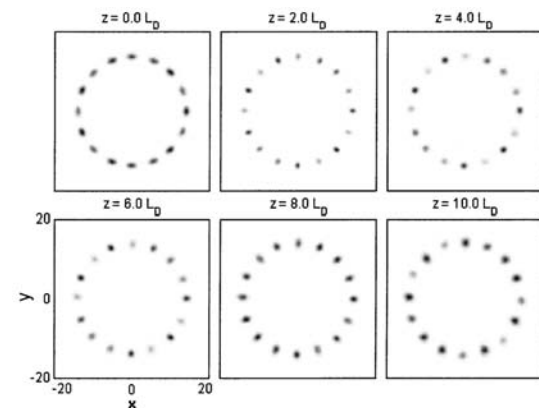


FIG. 4. A necklace with $L/E = -35/38$. This necklace is stable for $8L_D$. Necklaces with better stability are such that the energy exchange between the spots is slow, so it is not visible in a gray-level figure: for example, a necklace with $L/E = 261/1634$ that is stable for at least $50L_D$.

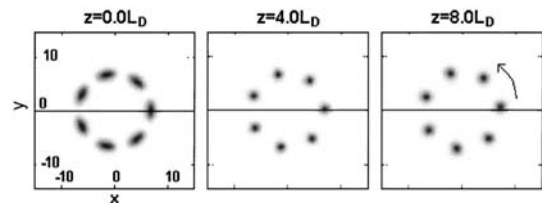


FIG. 3. A necklace with $L/E = 1/2$. This necklace slowly rotates as it propagates. The input shape is approximately $\psi(r, \theta, z = 0) = \text{sech}(r - 6.83) \{ \exp(i4\theta) + \exp(-i3\theta) \}/2$.

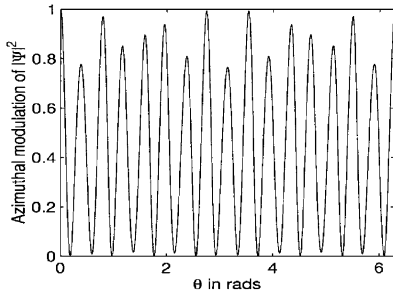


FIG. 5. Azimuthal intensity profile of the necklace of Fig. 4.

“rigid” as with the necklaces from Figs. 1 and 3, different parts of the necklace can have different angular velocities. Thus, the noninteger L/E of these beams does not imply that the angular momentum per photon is a noninteger multiple of $\hbar$. This is in contrast with the $M/2$ case, where the angular momentum per photon is $\hbar M/2$ everywhere.

Note that all necklaces described here are indeed self-trapped. If we start with a shape that is close to the equilibrium necklace shape, as long as this necklace is stable, both L and E are conserved: the L and E carried away by radiation are negligible compared to the initial L and E values. The self-trapped necklace conserves its initial L/E to even much better accuracy, although a tiny fraction of L and E is carried away through radiation.

To the best of our knowledge the necklaces described here are the only self-trapped shapes that have a non-integer per-photon angular momentum (in units of $\hbar$). Necklace beams can be constructed in many nonlinear wave equations. For example, one might think about converting the fractional angular momentum per particle carried by a necklace with $L/E = M/2$ into the angular momentum carried by the spin. This will imply rotation of the polarization state in optics, or spin-orbit interaction in a coherent system, such as a Bose-Einstein condensate. Such a conversion should be even more interesting when the necklaces are made of few photons only [21] (as opposed to a macroscopic number of photons [22]). Another exciting possibility is to investigate atomic necklaces in Bose-Einstein condensates.

We acknowledge enlightening discussions with Professor Meir Orenstein of Technion, Israel. This work was supported by the MURI Project on Optical Spatial Solitons.

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- [18] $f(r)$ must go at least as fast as r^2 close to the origin, so no singularities appear when this ψ is substituted into the NLSE; this issue is explained in [3,4].
- [19] The diffraction length L_D equals 1 normalized unit. If the nonlinearity is turned “off” in Eq. (1), then a beam of $\psi(x, z = 0) = \exp(-x^2/2)$ expands by $\sqrt{2}$ within $1L_D$.
- [20] The expectation value of the angular momentum of each photon is the same, anywhere on the necklace.
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